

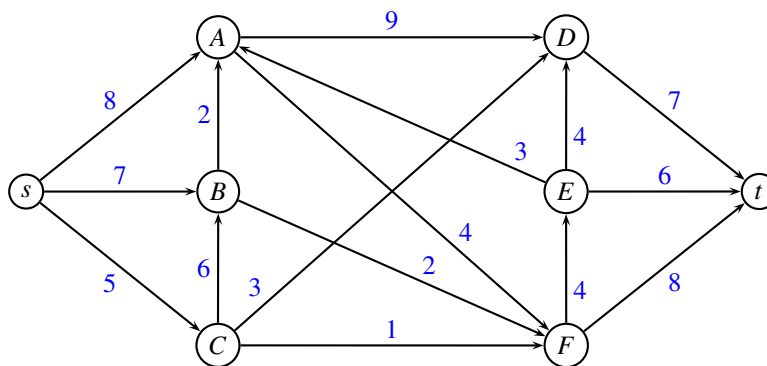
Answer, exam December, 2017.

Course name: Network Optimization

Course no: 42115

Maximum flow

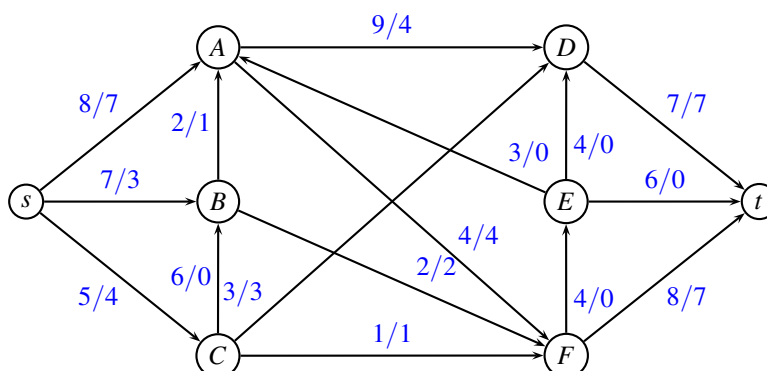
Consider the following maximum flow problem where edge weights denote the capacity of an edge.



Answer 1 We use the augmenting paths

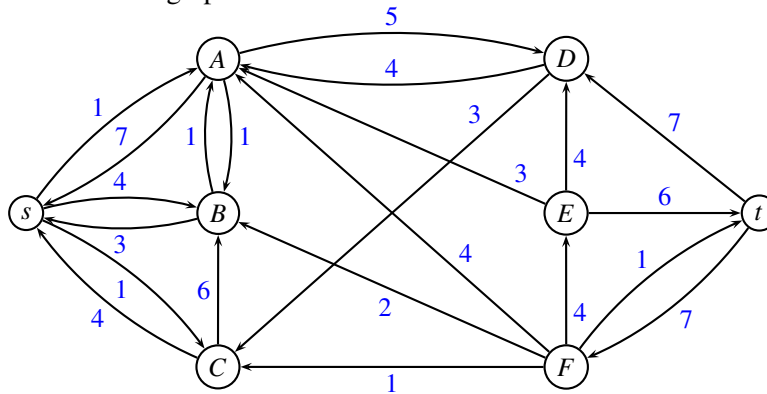
- $sADt$ capacity 7
- $sBFt$ capacity 2
- $sCFt$ capacity 1
- $sCDAFt$ capacity 3
- $sBAFt$ capacity 1

(There are many alternative ways of finding augmenting paths). This results in the flow



The maximum flow is 14. ■

Answer 2 We get the residual graph



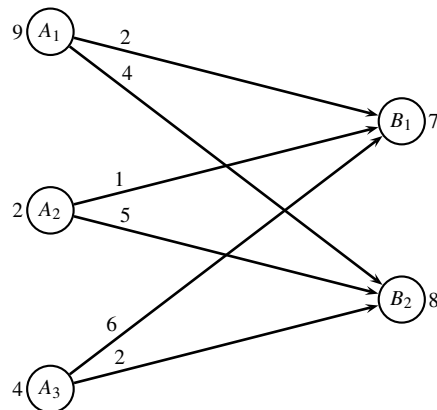
A minimal cut is $S = \{s, A, B, C, D\}$ and $T = \{E, F, t\}$. The capacity of the cut is 14. ■

Transportation problem

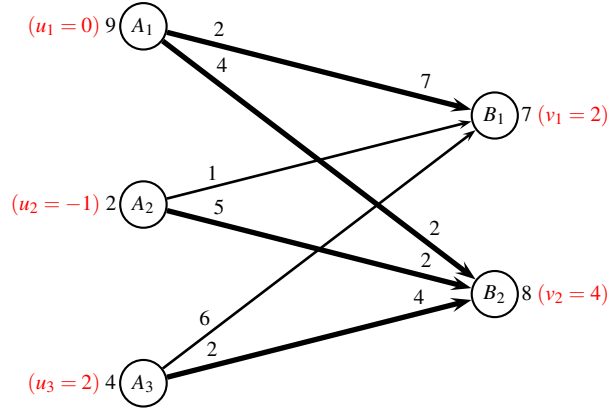
Consider the following transportation problem. We have three suppliers A_1, A_2, A_3 and two customers B_1, B_2 . The cost of transporting one unit from A_i to B_j is given in the following table

	B_1	B_2	supply
A_1	2	4	9
A_2	1	5	2
A_3	6	2	4
demand	7	8	

Answer 3 The problem can be illustrated as



We solve the problem with the greedy algorithm getting the solution



The cost of the solution is $2 \cdot 7 + 4 \cdot 2 + 5 \cdot 2 + 2 \cdot 4 = 40$.

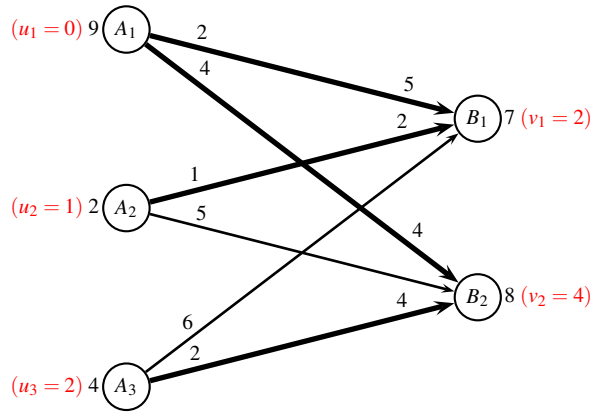
Fat edges on the graph indicate the basis. Dual values (indicated with red) are found by finding the shortest path from node A_1 , using only the fat edges.

We now find the reduced costs as $\bar{c}_{ij} = c_{ij} + u_i - v_j$, getting

$$\begin{aligned}\bar{c}_{21} &= 1 + (-1) - 2 = -2 \\ \bar{c}_{31} &= 6 + 2 - 2 = 6\end{aligned}$$

We choose edge (A_2, B_1) corresponding to the cycle (A_2, B_1, A_1, B_2) . We can send at most $\delta = \min\{7, 2\} = 2$ units along this cycle.

After the pivot operation we get the solution



The cost of the solution is $2 \cdot 5 + 4 \cdot 4 + 1 \cdot 2 + 2 \cdot 4 = 36$.

Dual values (indicated with red) are found by finding the shortest path from node A_1 , using only the fat edges.

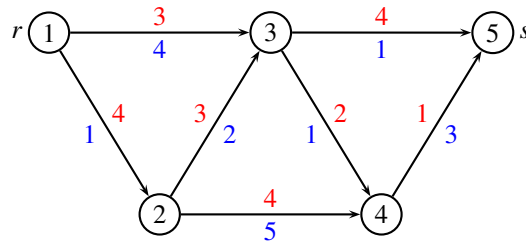
We now find the reduced costs as

$$\begin{aligned}\bar{c}_{22} &= 5 + 1 - 4 = 2 \\ \bar{c}_{31} &= 6 + 2 - 2 = 6\end{aligned}$$

Since no edges have negative reduced cost, the solution is optimal. ■

Resource constrained shortest path problem

Consider the following directed acyclic graph. Every edge (i, j) has an associated cost c_{ij} (red) and time t_{ij} (blue). We are searching for a shortest path (with respect to cost) from r to s . There is a time constraint T imposed on the path. We would like to find the shortest path for every relevant value of T .



Answer 4 First notice that the graph is a directed acyclic graph (DAG). We can therefore order the nodes in topological order, and use a label setting algorithm (i.e. dynamic programming) to solve the problem.

It is also OK to use a label correcting algorithm, but we only need to consider each node once, so it is not necessary to have a queue Q . ■

Answer 5 We consider the nodes in order 1,2,3,4,5 finding the lists

node 1	node 2	node 3	node 4	node 5
cost time	cost time	cost time	cost time	cost time
0 0	4 1	7 3	9 4	11 4
		3 4	5 5	7 5
			(8 6)	6 8
				(10 7)

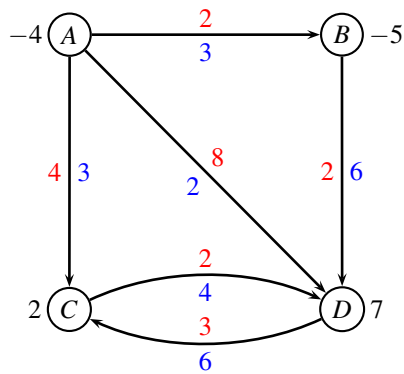
The dominated entries are marked with a paranthesis. This means that we can reach node s with the following costs and times:

cost	time
11	4
7	5
6	8

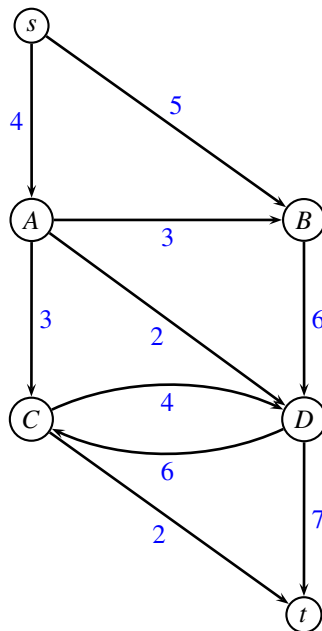
This means, for a time limit $T < 4$ we cannot find any solution. For time $T \geq 4$ cost is 11. For time $T \geq 5$ cost is 7. For time $T \geq 8$ cost is 6. ■

Min cost flow

Consider the following network with edge costs c_{ij} (red) and edge capacities u_{ij} (blue). The demand b_i (black) in every node is positive for a sink, and negative for a source.



Answer 6 We find an initial solution by solving a maximum flow problem

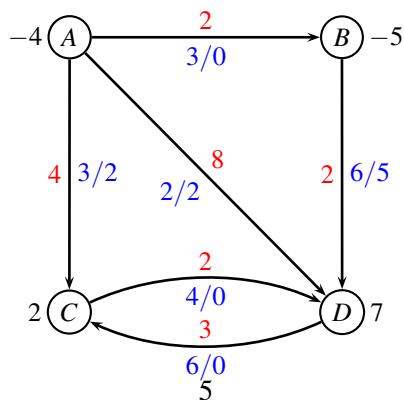


We use an augmenting path method, sending flow

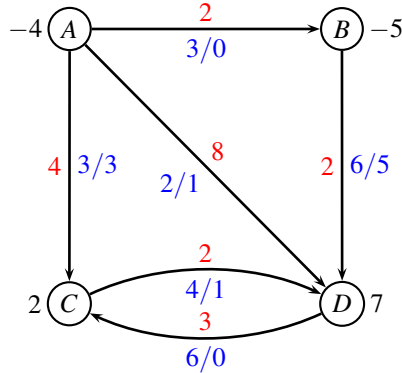
- $sACt$ flow 2
- $sADt$ flow 2
- $sBDt$ flow 5

The solution value is $4 \cdot 2 + 8 \cdot 2 + 2 \cdot 5 = 34$. ■

Answer 7 We have the initial solution

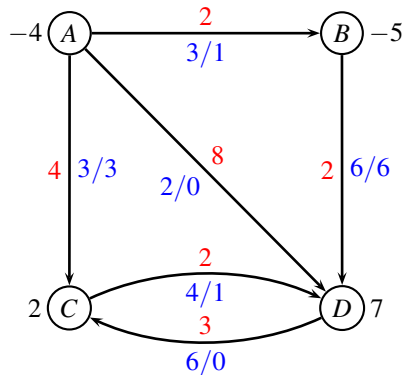


Using the augmenting cycle algorithm, we find a cycle $ACDA$ with cost $4 + 2 - 8 = -2$. The capacity we can send along the cycle is $\delta = \min\{1, 4, 2\} = 1$. Sending 1 unit along the cycle we get:



The solution value is $4 \cdot 3 + 8 \cdot 1 + 2 \cdot 5 + 2 \cdot 1 = 32$.

We find a new cycle $ABDA$ with cost $2 + 2 - 8 = -4$. The capacity we can send along the cycle is $\delta = \min\{3, 1, 1\} = 1$. Sending 1 unit along the cycle we get:



No more negative cycles can be found, so the solution is optimal. The solution value is $2 \cdot 1 + 4 \cdot 3 + 2 \cdot 6 + 2 \cdot 1 = 28$. ■

Critical path problem

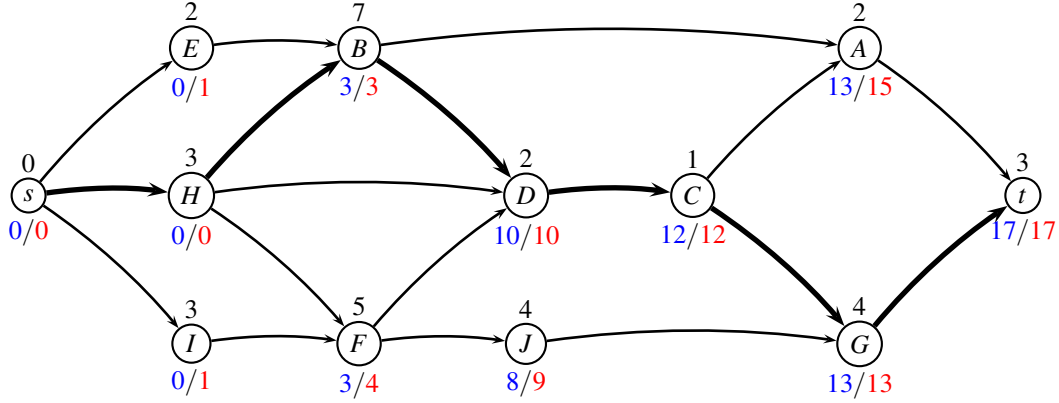
We have the activities

Activity	Description	Predecessors	Duration
A	Chain	B, C	2
B	Crankset	E, H	7
C	Chainring	D	1
D	Handlebar	B, F, H	2
E	Spindle		2
F	Wheel	H, I	5
G	Pedals	C, J	4
H	Fork		3
I	Rear gear		3
J	Brakes	F	4

Answer 8 We order the tasks in topological order, getting the ordering

$$E, H, I, B, F, D, J, D, A, G$$

which also can be drawn as follows:



In this graph, we inserted a new node s and connect it to the three activities E, H, I with no predecessors. Moreover, we inserted a new node t and connect it to the two activities A, G with no successors.

■

Answer 9 Now we solve the problem

- *Forward procedure:* Starting at time zero, calculate the earliest time U_j each job j can be started $U_j = \max_{(i,j) \in E} \{U_i + p_i\}$. (marked with blue, above)
- *Makespan:* $T = U_t = 17$.
- *Backward procedure:* Starting at time equal to the makespan, calculate the latest time V_j each job j can be started so that this makespan is realized. $V_i = \min_{(i,j) \in E} \{V_j - p_i\}$ (marked with red, above)

■

Answer 10 The critical path is marked with fat edges. There is only one critical path. Although both node H and D are critical, the edge (H, D) is not on the critical path. ■

Arranging a dinner

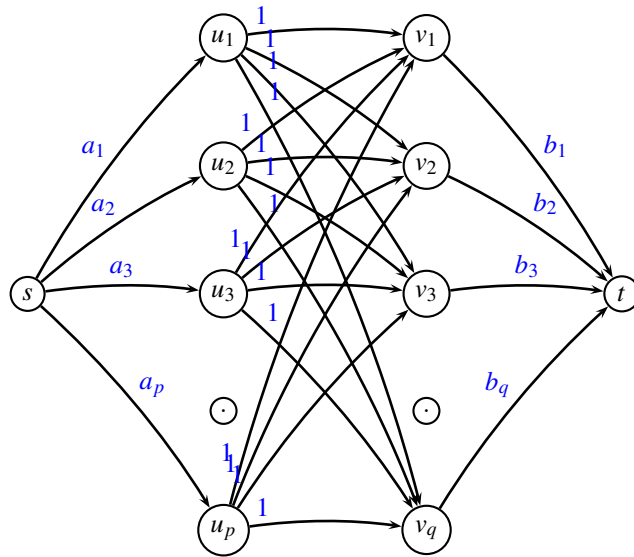
Answer 11 We define a capacitated graph $G = (V, E)$ as follows

$$V = \{s, t\} \cup \{u_i, 1 \leq i \leq p\} \cup \{v_j, 1 \leq j \leq q\}$$

and

$$E = \{(s, u_i), 1 \leq i \leq p\} \cup \{(u_i, v_j), 1 \leq i \leq p, 1 \leq j \leq q\} \cup \{(v_j, t), 1 \leq j \leq q\}$$

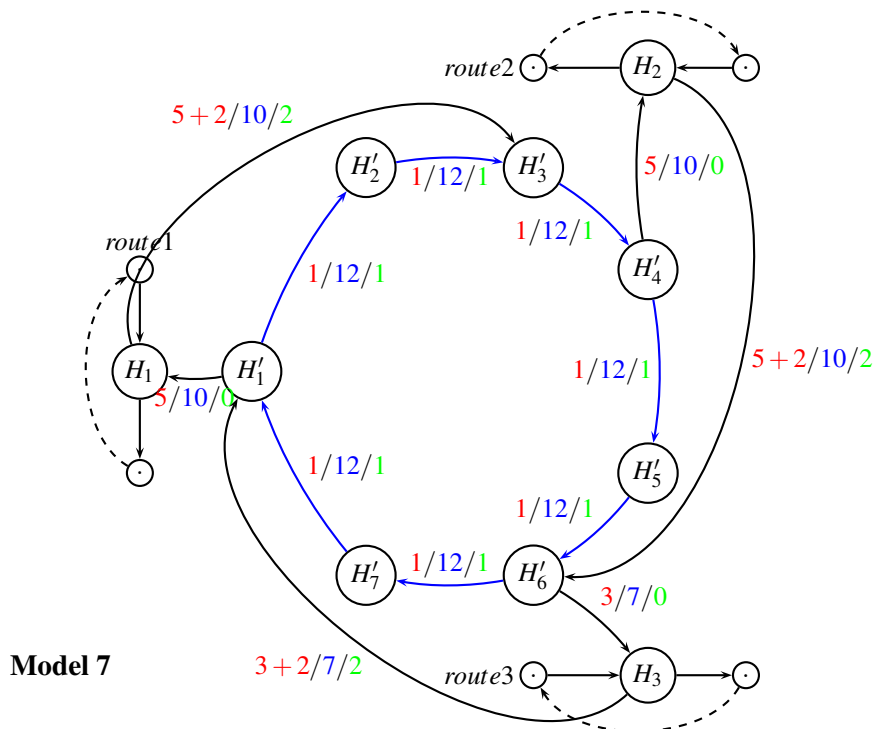
The capacity of edges (s, u_i) is set to a_i , the capacity of edges (u_i, v_j) is set to 1, and the capacity of edges (v_j, t) is set to b_j .



An integral solution to the maximum flow problem gives an assignment of families to tables. A flow from u_i to v_j is either zero or one, with one meaning that a member of the family i sits at table j . The maximum flow maximizes the number of people that can be seated under the stated constraints. ■

Modeling a hub

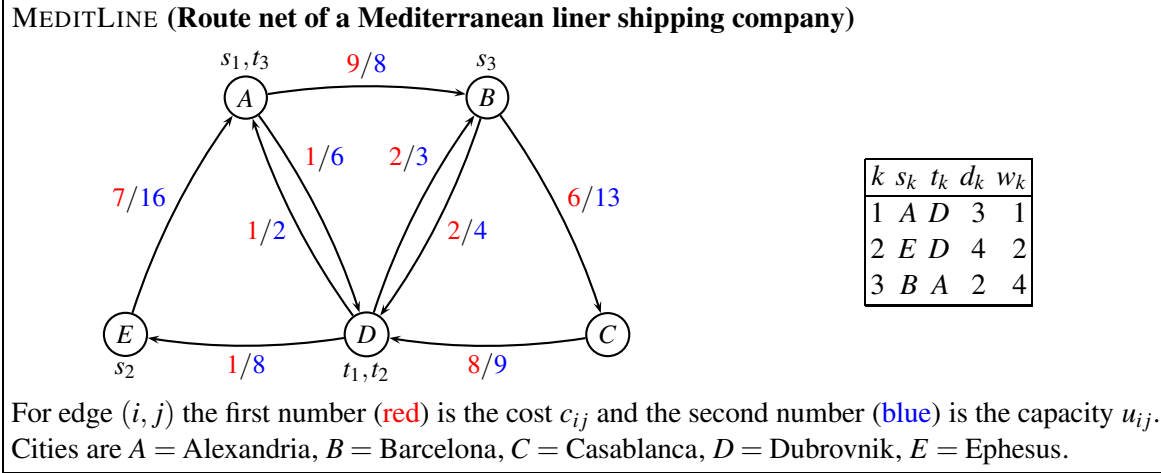
Answer 12 The loading and unloading edges are connected to the week day when a container is available after custom declaration.



In this formulation it is possible to handle transshipment cost and time, as well as crane capacity and storage cost and capacity (i.e. all the given data). Costs are indicated with **red**, capacities with **blue**, and time with **green** in the above figure.

Transshipment costs and crane capacity are assigned to edges between H_i and H'_i . Since the custom has infinite capacity for storing containers in a separate stock pile, it is only the crane capacity we need to respect on the unloading edges. Since it costs 1\$ per day the containers are stored in the custom, this cost has to be added to the cost of unloading edges. The capacity of edges between H'_i and H'_j is set to the storage capacity $\bar{c}$. Transshipment times are easily handled since every blue edge takes one day, loading edges take 0 days, and unloading edges take 2 days. ■

Flowing containers in a network



The dual variables are $y_{BD} = -12$, $\bar{y}_1 = 1$, $\bar{y}_2 = 100$, $\bar{y}_3 = 88$. All other dual variables are zero.

Answer 13 The reduced cost of a route r is calculated as

$$\bar{c}_r^k = \sum_{(i,j) \in r} w_k c_{ij} - \sum_{(i,j) \in r} w_k y_{ij} - \bar{y}_k$$

This leads to the following table:

number	commodity k	route r	w_k	$\sum w_k c_{ij}$	$\sum w_k y_{ij}$	$\bar{y}_k$	reduced cost $\bar{c}_r^k$
1	1	ABCD	1	23	0	1	22
2	1	ABD	1	11	-12	1	22
3	1	AD	1	1	0	1	0
4	2	EABCD	2	60	0	100	-40
5	2	EABD	2	36	-24	100	-40
6	2	EAD	2	16	0	100	-84
7	3	BCDEA	4	88	0	88	0
8	3	BDEA	4	40	-48	88	0

The most promising route to add to the model is route number 6 with reduced cost -84. ■